

The solar credit, dollar by dollar

Walking through a cost-benefit analysis of the Renewable Energy Technologies Income Tax Credit, and what the exercise teaches about evaluating any tax credit.

Dylan T. Moore, UHERO

Prepared with extensive research assistance from Branden Hing.

A tax credit is spending that happens to run through the tax code

Every dollar the credit gives back is a dollar that could have paid for anything else. So we ask the same four questions we would ask of any appropriation.

- 1 What does it cost the state, net?
- 2 What does it cause that would not have happened anyway?
- 3 Who ends up with the money?
- 4 What comes back per dollar spent?

\$106M

claimed under the credit in 2022.

\$49M

of that on corporate returns. Today's analysis covers the household half: residential solar claimed on individual returns.

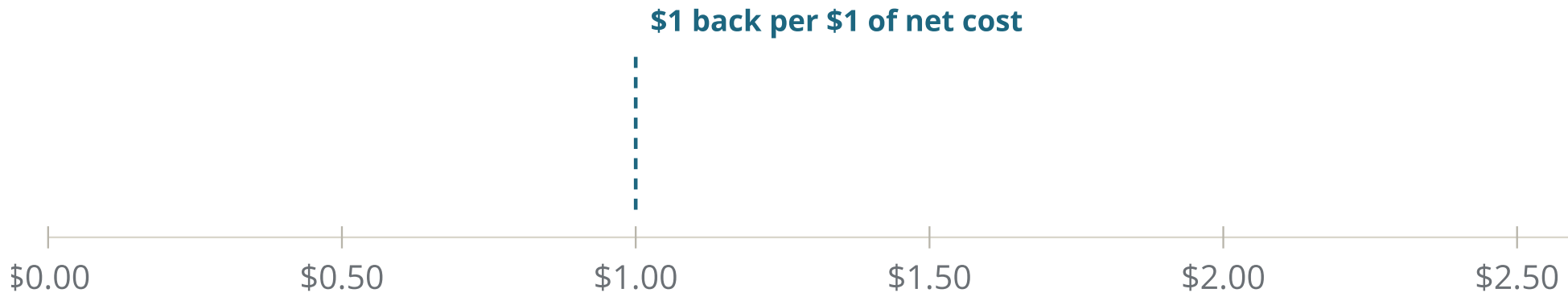
WHAT GENERALIZES

A credit is evaluated the same way as a spending program. The film credit, the solar credit, and a line item in the budget all compete for the same dollar.

For every dollar the state gives up, how much benefit comes back, and to whom?

The question is just: how much bang for the buck? We trace each dollar of net cost to the people it ends up helping or hurting, then add it up. Economists call the result the marginal value of public funds. Think of it as benefits per dollar.

The natural benchmark is **\$1.00**: the value of simply mailing residents a check. A program that scores above one does something a check cannot. One that scores below it is doing something a check would do better.



WHAT GENERALIZES

One number puts a tax credit, a grant, and a direct appropriation on the same footing. It makes "is this credit worth it?" a comparison rather than a verdict.

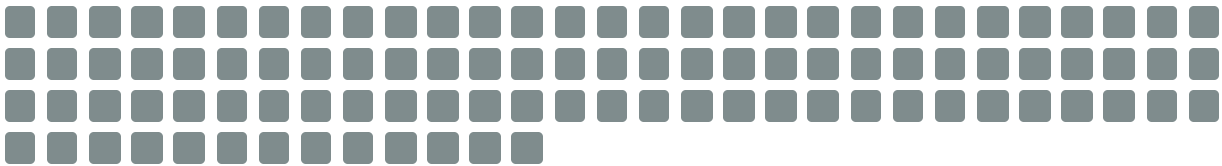
Who claims the credit? Two very different kinds of household

Picture 100 households that would have put panels on the roof even with no credit at all. With the credit, our best estimate is that about **45 more** join them.

So of every 145 households that claim the credit, 100 were coming anyway. The credit changed the behavior of the other 45.

 Would have installed anyway  Convinced by the credit

The 35% credit cuts the household's share of the cost to about 65 cents on the dollar.



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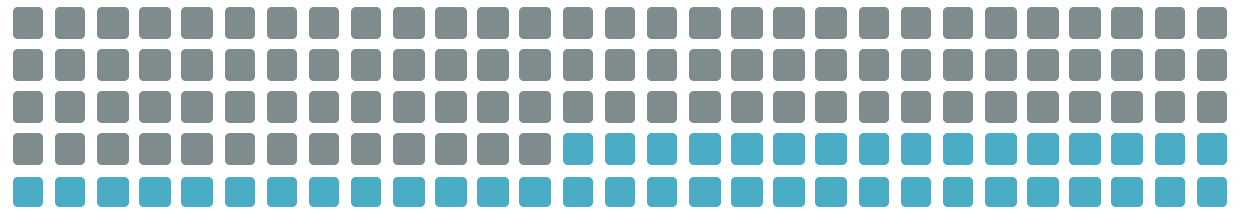
The first question for any credit is what it actually causes that would not have happened anyway. Money that goes to activity that was coming regardless is a transfer, not an incentive.

Film credit: how many productions would have shot here regardless? That is the first thing to find out, and the hardest.

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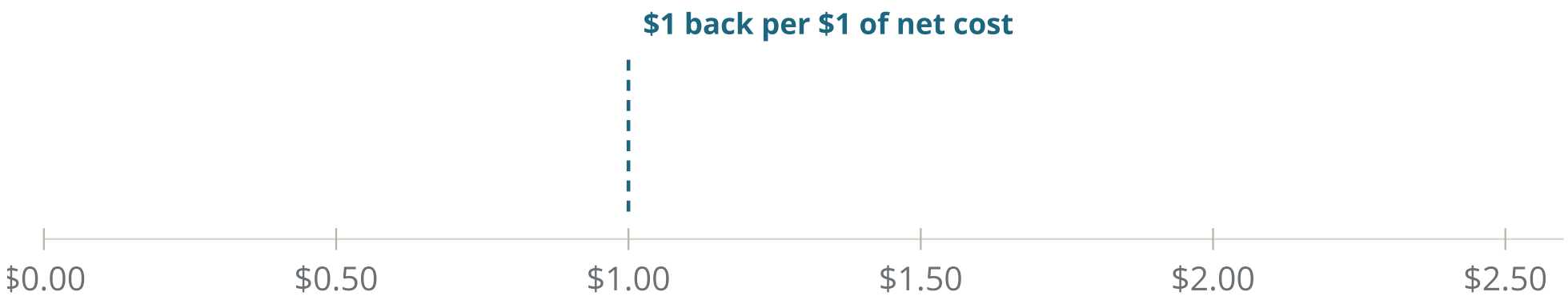
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Households that would have installed anyway: a check in the mail

For these households the credit does not change what they buy. It just makes them richer by the amount of the credit.

Economically that is worth exactly what it costs: a dollar to them is a dollar. Of every dollar of net cost, about **56 cents** lands here.

If the goal was more solar, the money spent on these households is, to a first approximation, lost.



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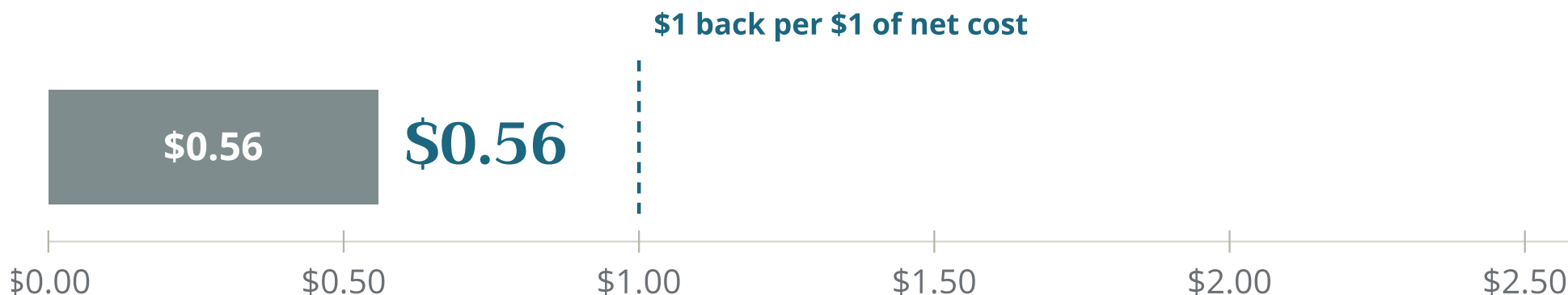
The largest slice of most incentive credits is a transfer to people who were going to do the thing anyway. Valued at par, it buys no new activity. It is the first thing to measure.

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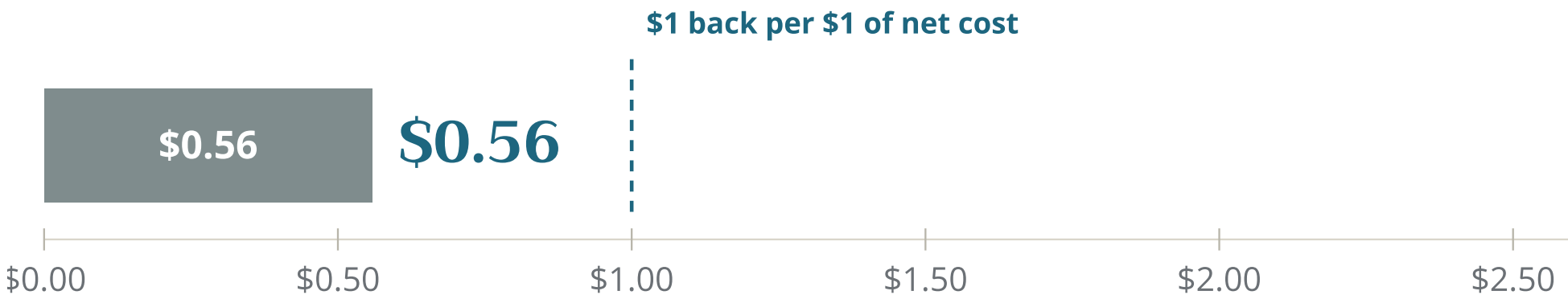
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Households the credit convinced: worth less than what we paid

Think about the very last household we convinced. They were on the fence. The credit tipped them, which means they value the panels at just about what they paid, and the credit itself at close to nothing.

Averaged over all the convinced households, the value they actually get from the credit comes to about **13 cents** per dollar of state spending.



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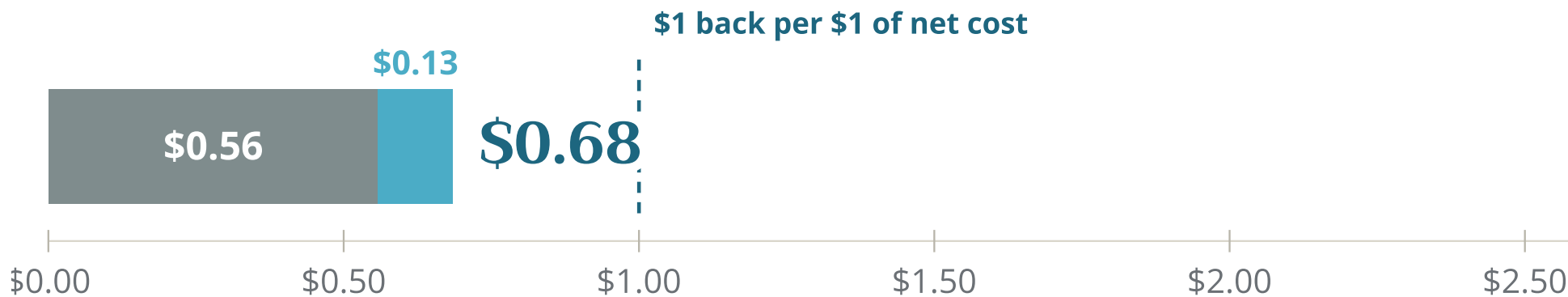
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This is why film-credit claims of '\$X in new spending' are not benefits. The productions the credit attracted are, by definition, the ones that were nearly indifferent about coming.

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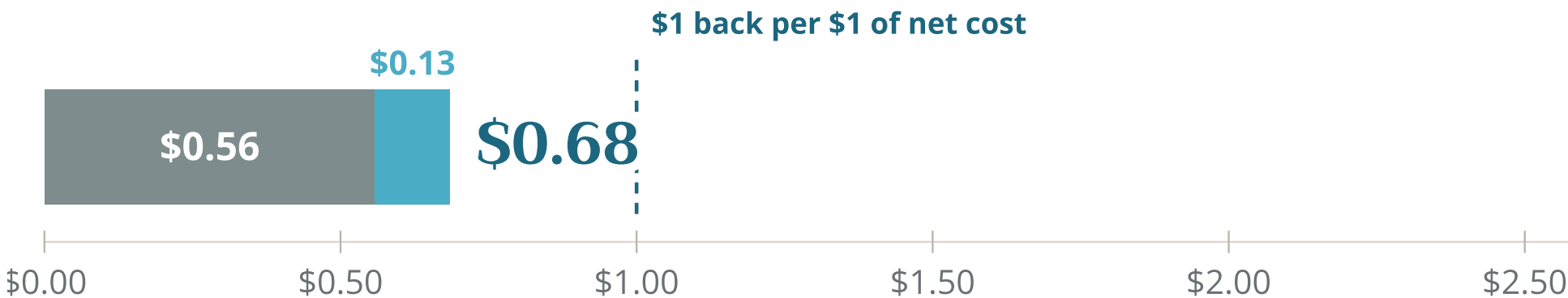
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Solar installers: when demand goes up, prices go up

The credit pushes more households into the market, and installers can charge higher prices. So part of the credit never reaches the household at all.

That shows up as higher profits for those firms, and perhaps a small increase in their workers' wages. We put it at about **16 cents** per dollar of state spending.



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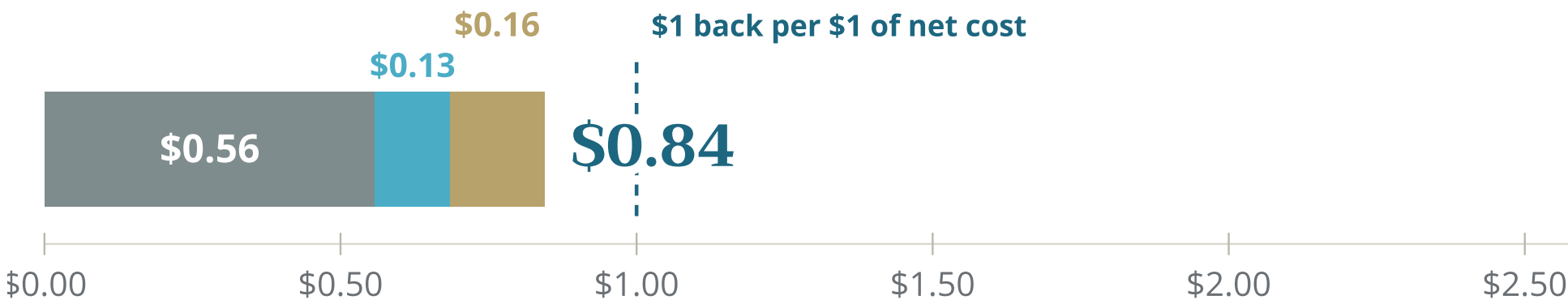
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The climate benefit: the reason the credit exists

Each of the 45 convinced households displaces electricity Hawai'i would otherwise make by burning oil. At the EPA's social cost of carbon, that is worth **\$1.70** per dollar of net cost, the biggest slice by far.

Most of that value is harm avoided for people outside Hawai'i. Climate change is global, so most of the benefit of fixing it is too.

If HECO delivers 60% of its planned grid decarbonization, this falls to \$0.97, because the displaced power gets cleaner.



WHAT GENERALIZES

Whose benefits count is a policy choice, not a technical one. Here it moves the answer from \$2.41 to \$0.71. The framework makes that choice visible; it does not make it.

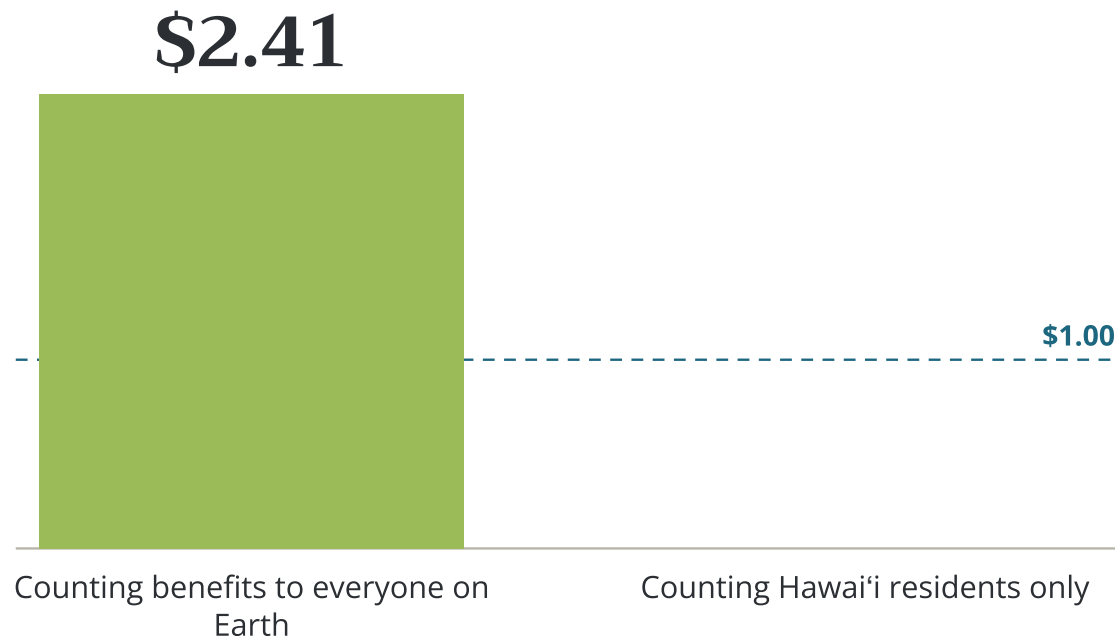
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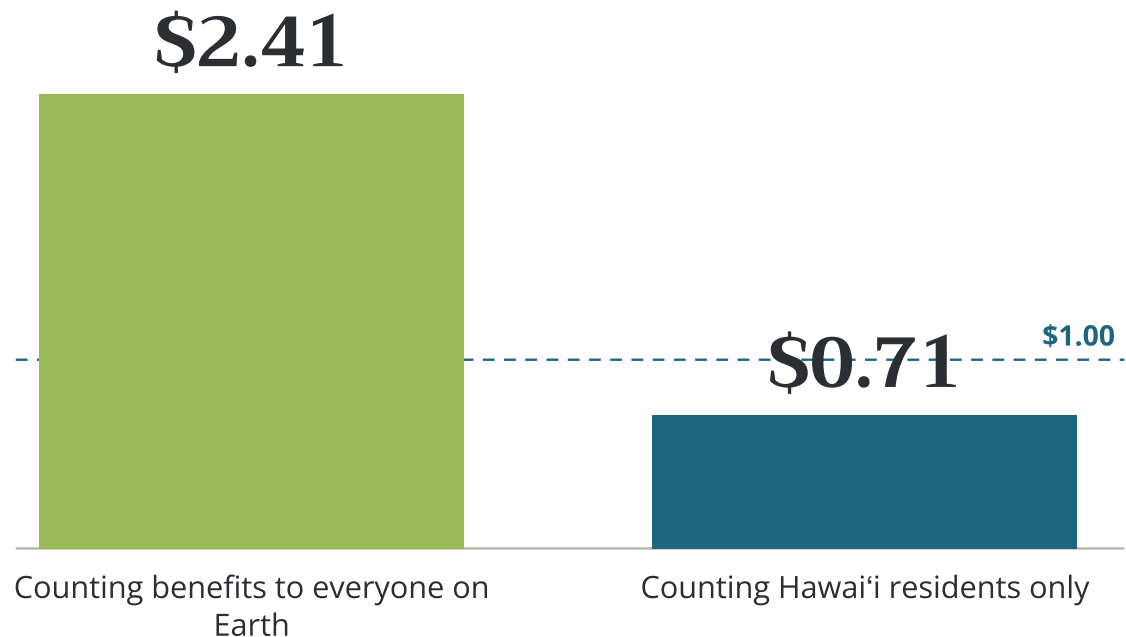
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What the climate number leaves out: local air quality

Burning less oil in Hawai'i also means less local air pollution. That benefit to residents is not in the \$1.70, and it proved difficult to estimate.

By some measures, local air quality has very little to do with the output of these power plants. The trade winds carry most of what the plants emit out over open ocean, where nobody lives.

Small

Our best judgment is that counting local air quality would raise the benefit to residents only modestly.

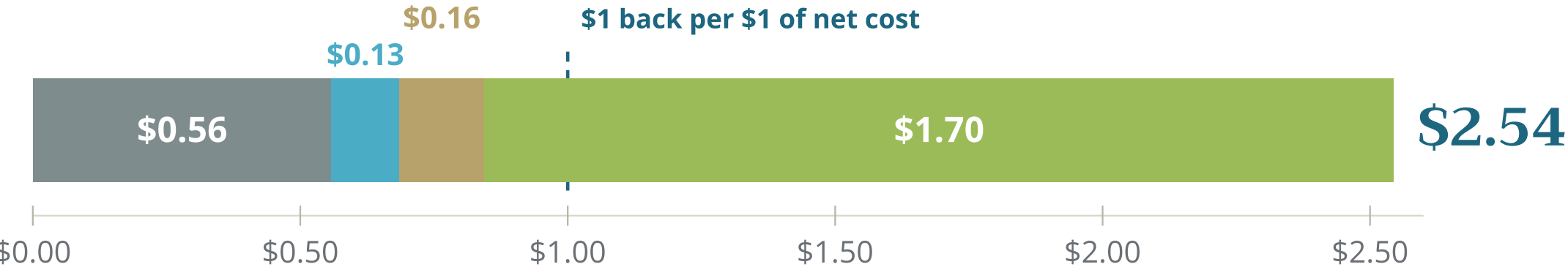
WHAT GENERALIZES

Not every benefit that sounds important turns out to be large or measurable. An evaluation says which ones are left out and which direction they would push the answer.

Other ratepayers: the bill for someone else's panels

Every new rooftop system means less power sold by the utility, which reduces its profits. HECO likely recovers a large portion of that through rate hikes.

Our preliminary estimate is about **14 cents** of cost per dollar of state spending, falling on some combination of the utility and all other residents' electricity bills.



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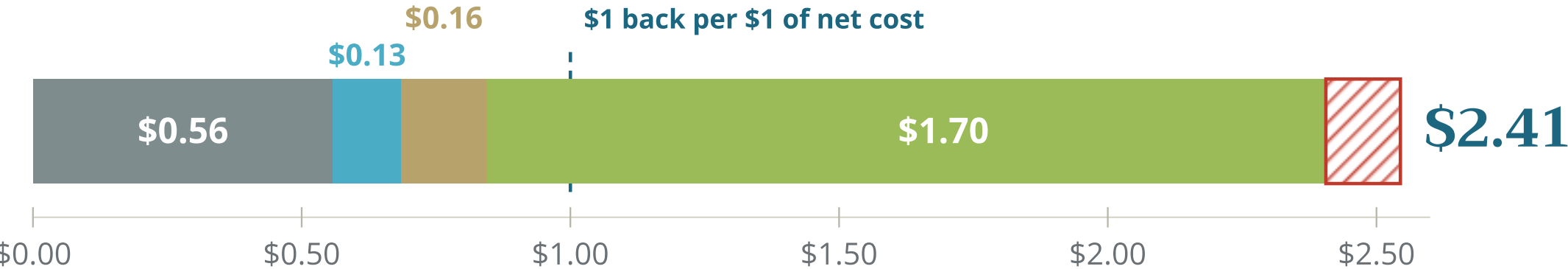
Costs that fall on people who never touch the program belong in the ledger. They are easy to leave out because nobody sends the state a bill for them.

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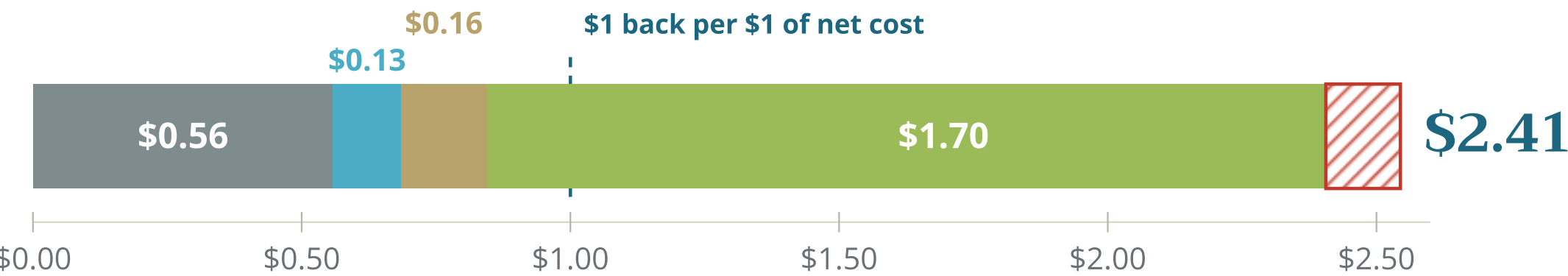
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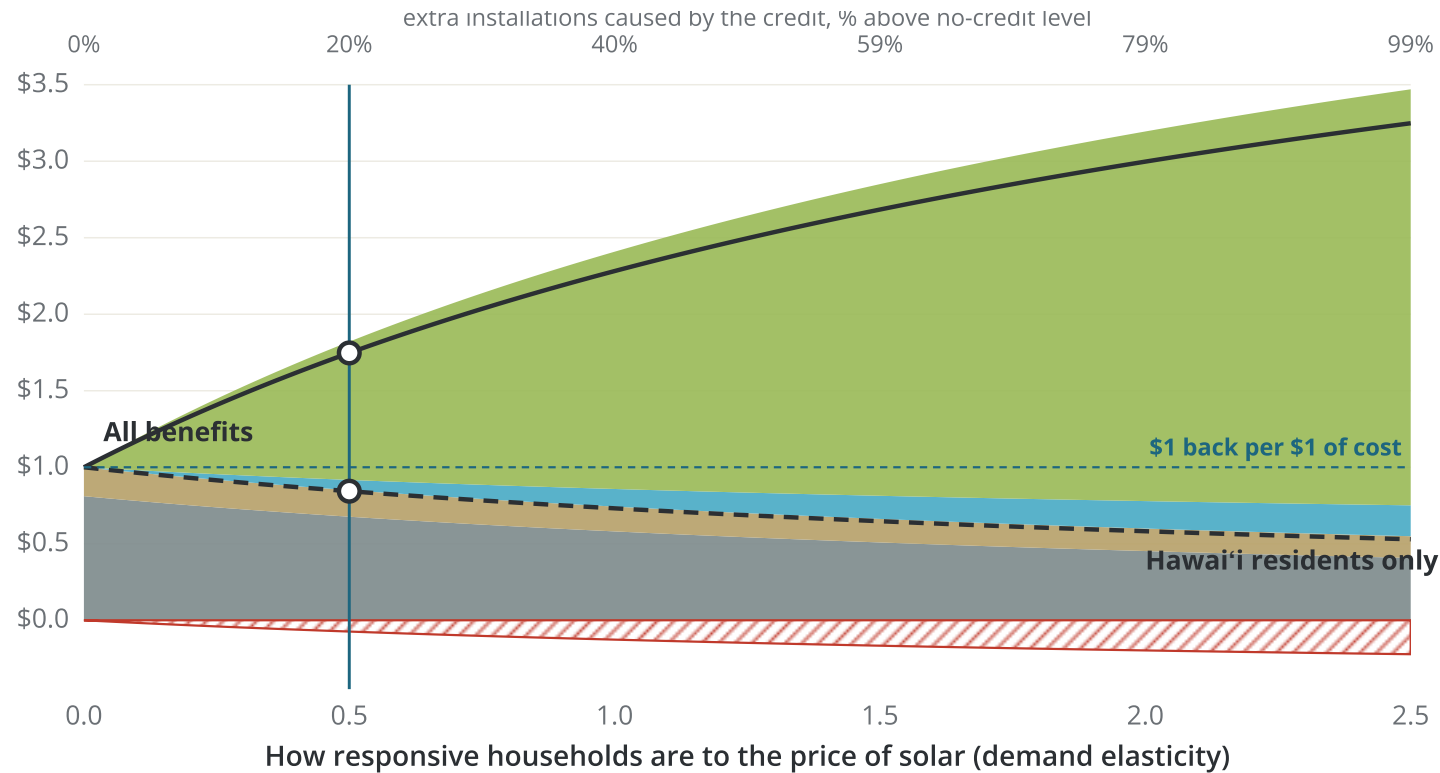
Adding it up: \$2.41 back per dollar, or \$0.71, depending on who counts

PER \$1 OF NET STATE COST	AMOUNT
Households that would have installed anyway	\$0.56
Households the credit convinced	\$0.13
Solar installers	\$0.16
Other ratepayers and the utility	-\$0.14
Hawai'i residents, total	\$0.71
Climate benefit, everyone on Earth	\$1.70
Everything, total	\$2.41

Counting residents only, the credit returns less than the dollar it costs, and most of what residents get is a transfer to households that were buying solar anyway.



How much do households respond to price? We borrowed the answer



Responsiveness assumed on this page

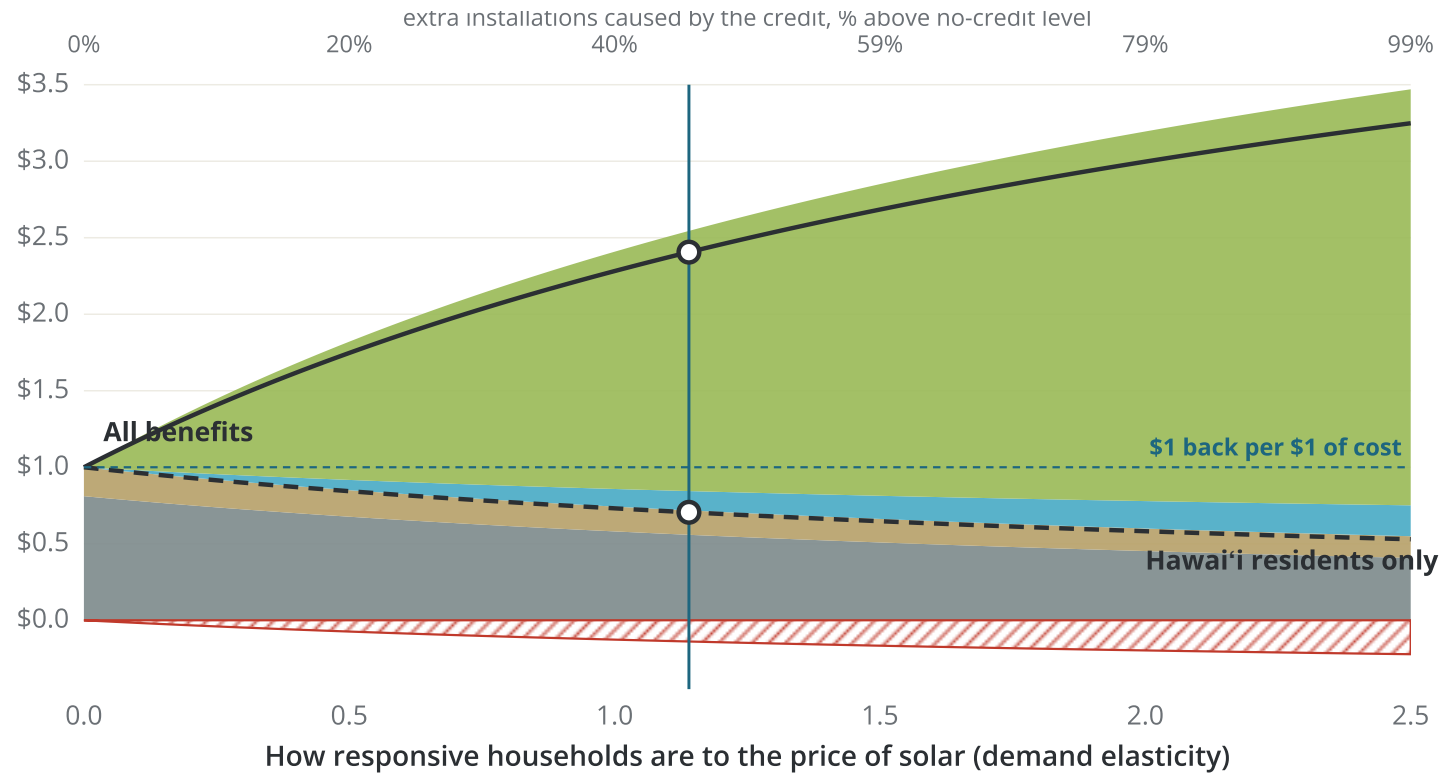
- 0.50 elasticity
- 20% extra installations caused by the credit
- \$1.75 back per dollar, everyone counted
- \$0.84 back per dollar, residents only
- \$339 state cost per ton of CO₂ avoided

WHAT GENERALIZES

The climate number rises and falls with this one parameter. The resident-only number barely moves. An honest evaluation of any credit shows a range like this, not a single number.

Hawai'i households could be less responsive than mainland ones, or more.

How much do households respond to price? We borrowed the answer



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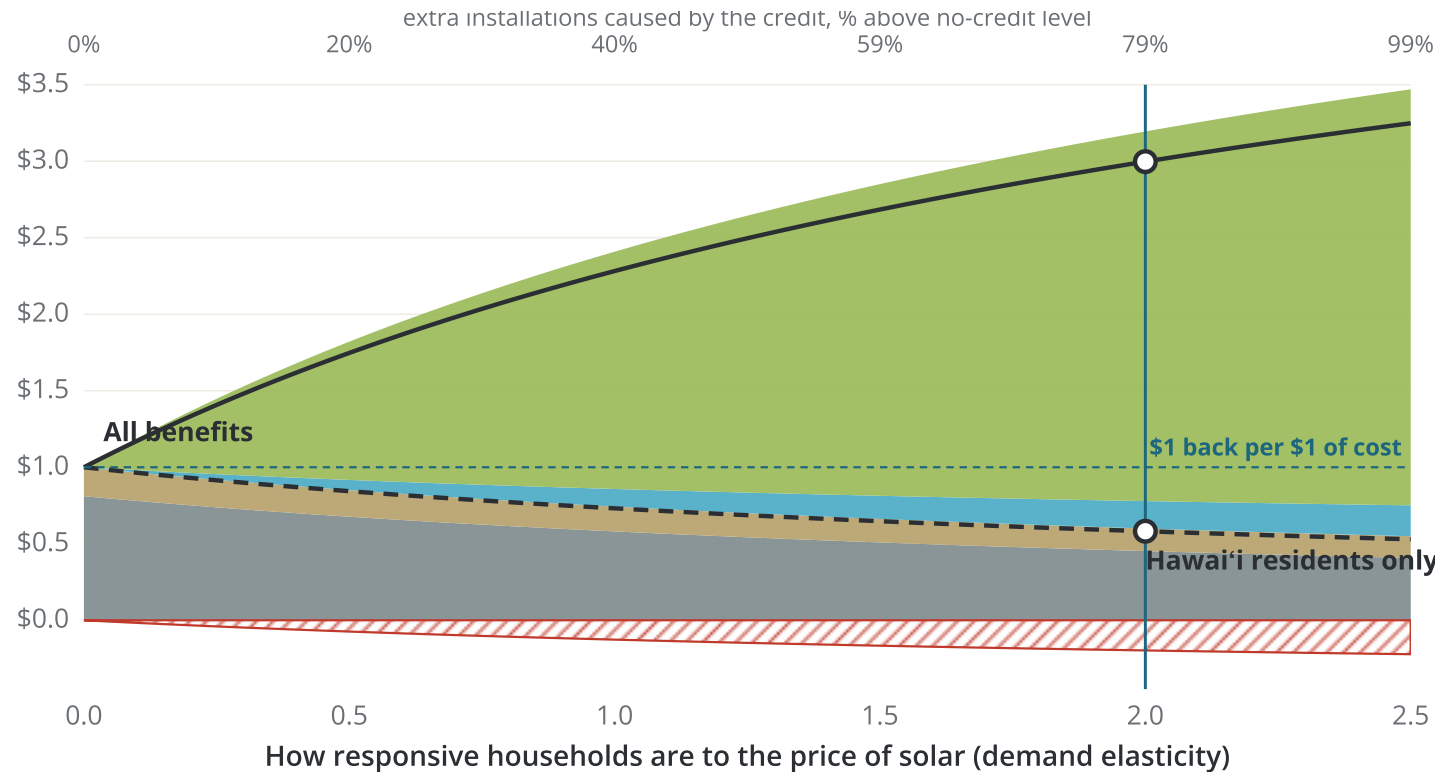
- 1.14** elasticity (the mainland estimate)
- 45%** extra installations caused by the credit
- \$2.41** back per dollar, everyone counted
- \$0.71** back per dollar, residents only
- \$180** state cost per ton of CO₂ avoided

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How much do households respond to price? We borrowed the answer



Responsiveness assumed on this page

- 2.00 elasticity
- 79% extra installations caused by the credit
- \$3.00 back per dollar, everyone counted
- \$0.58 back per dollar, residents only
- \$127 state cost per ton of CO₂ avoided

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On the mainland, utility-scale solar subsidies return 2 to 3 times as much per dollar

2–3×

the benefits per dollar of net state cost, relative to residential solar subsidies, in a recent study of mainland programs.

Utility-scale generation is simply cheaper per kilowatt-hour. That makes a utility-scale subsidy worth considering in Hawai'i.

The mainland estimates may not carry over directly, and no analysis of such a program has been done for Hawai'i, so the size of the gain here is not known.

Its benefits could also be spread more evenly across residents, provided the cost savings are passed through to rates.

WHAT GENERALIZES

The question is never whether a credit has benefits. It is whether the same dollar buys more of the goal some other way. An alternative belongs next to every credit.

For the film credit: what would a direct grant of the same size buy, and could it be aimed at local firms and workers directly?

The credit flows mostly to higher-income households

Median income and above

is where most claimants sit.

Earlier UHERO work shows that high-income households are the ones most able to benefit from the credit. Mainland research finds the same for similar programs elsewhere.

Claiming the credit takes a roof the household owns and enough tax liability to use a nonrefundable credit. Renters and lower-income households are largely left out.

The cost side is spread more widely: rate hikes reach every ratepayer, including those who never claimed the credit.

WHAT GENERALIZES

Who benefits is a separate question from whether a credit is efficient. Both are reported, and each is weighed on its own.

For the film credit: which residents, and which businesses, end up with the money?

Eight questions to ask of any tax credit

- 1 Net cost.** Credits paid, plus revenue lost or gained elsewhere because of the activity.
- 2 What it causes.** How much of the activity would have happened without it?
- 3 Transfers.** Dollars to people who were doing it anyway are a transfer, valued at par, that buys nothing new.
- 4 Induced activity.** Worth less than it cost to induce. Only external benefits can make up the gap.
- 5 Incidence.** Who actually ends up with the money once prices adjust?
- 6 Third parties.** Costs and benefits to people who never touch the program.
- 7 Whose benefits.** Residents only, or everyone? Both are reported.
- 8 The alternative.** Cost per unit of the goal, next to the cheapest other way to get it.